

Marketing Analytics MKT - 568

Final Project

By

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METHODOLOGY

The analysis began by importing and inspecting the IMDb movie dataset, which originally contained 5,438 observations and 15 variables. Initial data exploration included examining variable types, identifying missing values, checking duplicates, and reviewing descriptive information using `.info()`, `.describe()`, and `.isnull().sum()`. After inspection, rows with missing values in *rating*, *company*, *country*, and *runtime* were removed because the number of missing entries was small and deletion would not significantly reduce the dataset. Removing these rows resulted in a cleaned dataset of 5,423 movies.

To prepare the data for analysis, several new variables were engineered. The *release* column was cleaned by removing country tags such as “(United States)” and converting the remaining text into a proper datetime format. From this cleaned release date, new variables were extracted: **release_year**, **release_month**, **release_day**, and **release_season**. The **release_day** variable was further transformed into **release_day_bin**, using the required intervals (1–5, 6–10, 11–15, 16–20, 21–25, 26–31). Financial variables were enhanced by adjusting the production budget for inflation using the average annual inflation rate of 3.09%, creating **budget_inflation_adjusted**. Profit was then calculated as **gross – adjusted budget**. Additional engineered variables included **movie_name_length** and **years_since_production**, both of which support later predictive modeling.

Outliers were addressed using Winsorization at the 1st and 99th percentiles for the key continuous variables (budget, gross, profit, runtime, score, votes, years_since_production). This approach caps extreme values without discarding observations, ensuring robustness while preserving data integrity. Categorical variables—including rating, genre, company, release_season, and release_day_bin—were dummy coded using one-hot encoding with `drop_first=True` to prevent multicollinearity. Director, writer, star, and country were excluded from dummy coding per project instructions.

For documentation and modeling clarity, a full measurement table was created to classify each variable as categorical or continuous. Summary statistics were generated for all continuous variables, and a correlation matrix was constructed to examine linear relationships among budget, revenue, profit, score, runtime, votes, and other numerical variables. These steps completed the data cleaning and preparation process, ensuring that the dataset was

accurate, consistent, and ready for descriptive, inferential, and predictive analysis in subsequent sections.

PART A: DATA CLEANING & PREPARATION

A1. Introduction to the Dataset

The dataset includes 5,423 movies collected from IMDb and contains variables such as title, rating, genre, release information, budget, gross revenue, production company, and runtime. Several new variables were created for analysis, and the dataset was cleaned to ensure accuracy and consistency before conducting descriptive, inferential, and predictive modeling.

A2. Variable Creation

Multiple new variables were created based on the released date and financial fields to support later analysis.

Release-Based Variables

The released column was cleaned to remove country tags (e.g., “(United States)”) and standardized to proper datetime format. From this cleaned date, the following variables were extracted:

- **release_year** – the year of release
- **release_month** – month name (e.g., January, March)
- **release_day** – numerical day of the month
- **release_season** – categorized into Spring, Summer, Fall, Winter

Release Day Binning

The release day was converted into categorical bins using the required intervals: **1–5, 6–10, 11–15, 16–20, 21–25, 26–31**, forming the variable **release_day_bin**.

Inflation Adjustment

The original budget values were adjusted to 2025 dollars using the average annual inflation rate of 3.09%.

The formula applied was:

$$\text{budget_inflation_adjusted} = \text{budget} \times (1 + 0.0309)^{(2025 - \text{year})}$$

Profit Calculation

Profit was computed as:

$$\text{profit} = \text{gross} - \text{budget_inflation_adjusted}$$

Additional Engineered Variables

- **movie_name_length** – number of characters in the movie title
- **years_since_production** – years passed since release year (2025 – release_year)

A3. Dummy Coding of Categorical Variables

Categorical variables (rating, genre, company, release_season, release_day_bin) were dummy coded, excluding director, writer, star, and country as required.

Dummy variables were created with **drop_first=True** to avoid multicollinearity.

The resulting dataset expanded from 15 to **1,526 columns**, reflecting the additional encoded variables.

A4. Handling Missing Values

The dataset originally contained missing values in four columns (rating, company, country, runtime).

Because the proportion of missing entries was very small, rows containing missing values in these variables were removed using:

```
df.dropna(subset=['rating', 'company', 'country', 'runtime'], inplace=True)
```

After removal, the dataset size became **5,423 observations**, ensuring high reliability while preserving almost all data.

A5. Outlier Treatment

Outliers were handled using **Winsorization (capping)** at the 1st and 99th percentiles for the following continuous variables:

- budget_inflation_adjusted
- gross
- profit
- votes
- runtime
- score
- years_since_production

This method preserves all observations but prevents extreme values from distorting results.

A6. Measurement Table

The table below identifies each variable and its measurement type (categorical or continuous), including newly created fields.

Table A1. Variable Measurement Types

Index	Variable Name	Measurement Type
0	name	Categorical
1	year	Continuous
2	released	Categorical
3	score	Continuous
4	votes	Continuous
5	director	Categorical
6	writer	Categorical
7	star	Categorical
8	country	Categorical
9	budget	Continuous
10	gross	Continuous
11	runtime	Continuous
12	release_date_clean	Date / Continuous
13	release_year	Continuous
14	release_month	Categorical
15	release_day	Continuous
16	budget_inflation_adjusted	Continuous
17	profit	Continuous
18	movie_name_length	Continuous

Index	Variable Name	Measurement Type
19	years_since_production	Continuous
20	rating_G	Categorical
21	rating_NC-17	Categorical
22	rating_Not Rated	Categorical
23	rating_PG	Categorical
24	rating_PG-13	Categorical
25	rating_R	Categorical
26	rating_TV-MA	Categorical
27	rating_Unrated	Categorical
28	rating_X	Categorical
29	genre_Adventure	Categorical

A7. Summary Statistics for Continuous Variables

Descriptive statistics were calculated for key continuous variables, including mean, median, standard deviation, and quartiles. These statistics summarize central tendencies and variability across financial and performance variables

Table A2. Summary Statistics (Continuous Variables)

	mean	median	std \
budget	35,997,734.86	21,500,000.00	41,570,081.03
budget_inflation_adjusted	63,031,790.34	46,105,045.38	53,916,641.71
gross	75,577,345.49	36,869,414.00	86,248,936.50
profit	12,549,940.21	-2,154,404.01	63,614,972.01
score	6.40	6.50	0.93
votes	89,836.02	52,000.00	92,681.70
runtime	107.73	105.00	16.56
movie_name_length	14.88	13.00	7.99
years_since_production	23.19	23.00	10.56

	min	25%	50% \
budget	3,000.00	10,000,000.00	21,500,000.00
budget_inflation_adjusted	10,770.01	22,061,245.64	46,105,045.38
gross	309.00	10,730,814.50	36,869,414.00
profit	-124,838,501.59	-25,605,979.05	-2,154,404.01
score	3.85	5.80	6.50
votes	195.00	19,000.00	52,000.00
runtime	63.00	95.00	105.00
movie_name_length	1.00	9.00	13.00
years_since_production	5.00	14.00	23.00

	75%	max
budget	45,000,000.00	356,000,000.00
budget_inflation_adjusted	88,313,462.51	187,691,787.81
gross	112,403,010.50	264,911,304.50
profit	40,549,035.98	139,781,558.52
score	7.10	9.05
votes	129,000.00	294,000.00
runtime	118.00	152.50
movie_name_length	18.00	83.00
years_since_production	32.00	45.00

A8. Correlation Matrix for Continuous Variables

A correlation table was generated to examine relationships between continuous variables such as budget, profit, gross revenue, score, runtime, and votes.

Table A3. Correlation Matrix

	budget	budget_inflation_adjusted	gross	profit	\
budget	1.00	0.89	0.71	0.30	
budget_inflation_adjusted	0.89	1.00	0.67	0.14	
gross	0.71	0.67	1.00	0.80	
profit	0.30	0.14	0.80	1.00	
score	0.07	0.03	0.22	0.28	
votes	0.51	0.45	0.73	0.63	
runtime	0.33	0.34	0.27	0.11	
movie_name_length	0.13	0.10	0.12	0.10	
years_since_production	-0.32	-0.10	-0.32	-0.36	

	score	votes	runtime	movie_name_length	\
budget	0.07	0.51	0.33	0.13	
budget_inflation_adjusted	0.03	0.45	0.34	0.10	
gross	0.22	0.73	0.27	0.12	
profit	0.28	0.63	0.11	0.10	
score	1.00	0.51	0.43	-0.02	
votes	0.51	1.00	0.35	0.04	
runtime	0.43	0.35	1.00	0.01	
movie_name_length	-0.02	0.04	0.01	1.00	
years_since_production	-0.06	-0.32	-0.09	0.01	

	years_since_production
budget	-0.32
budget_inflation_adjusted	-0.10
gross	-0.32
profit	-0.36
score	-0.06
votes	-0.32
runtime	-0.09
movie_name_length	0.01
years_since_production	1.00

Key observations:

- Profit and gross revenue show a strong positive correlation ($r \approx 0.80$).
- Votes and profit also show a high correlation ($r \approx 0.63$).
- Runtime and profit exhibit a very weak relationship ($r \approx 0.11$).

A9. Summary of Data Cleaning

A comprehensive cleaning process was conducted to prepare the dataset for analysis. Missing values were removed, outliers were winsorized, and all necessary variables were engineered and typed correctly. The cleaned dataset now contains **5,423 rows** and over **1,500 variables** after dummy encoding. These preprocessing steps ensure that the data is suitable for descriptive analytics, regression modeling, and machine learning tasks in subsequent sections.

PART B: DESCRIPTIVE ANALYSIS

Q1. Which companies have the highest average profitability?

To determine which production companies generate the highest profits, movies were grouped by company and their average profits were calculated. The top 10 companies all reported exceptionally high average profits, each reaching \$139,781,558.52. This outcome results from each company producing at least one highly profitable film within the dataset, demonstrating how blockbuster releases can significantly inflate a company's profitability.

Table 1. Top 10 Companies by Average Profit

Rank	Company	Average Profit (USD)
1	"Weathering With You" Film Partners	139,781,558.52
2	Waverly Films	139,781,558.52
3	888 Productions	139,781,558.52
4	Warner Bros. Pictures	139,781,558.52
5	Constantin Film International	139,781,558.52
6	Claudie Ossard Productions	139,781,558.52
7	Syncopy	139,781,558.52
8	Tokuma Shoten	139,781,558.52
9	Estudios Churubusco Azteca S.A.	139,781,558.52
10	The Mark Gordon Company	139,781,558.52

Q2. Which countries produce the most profitable films?

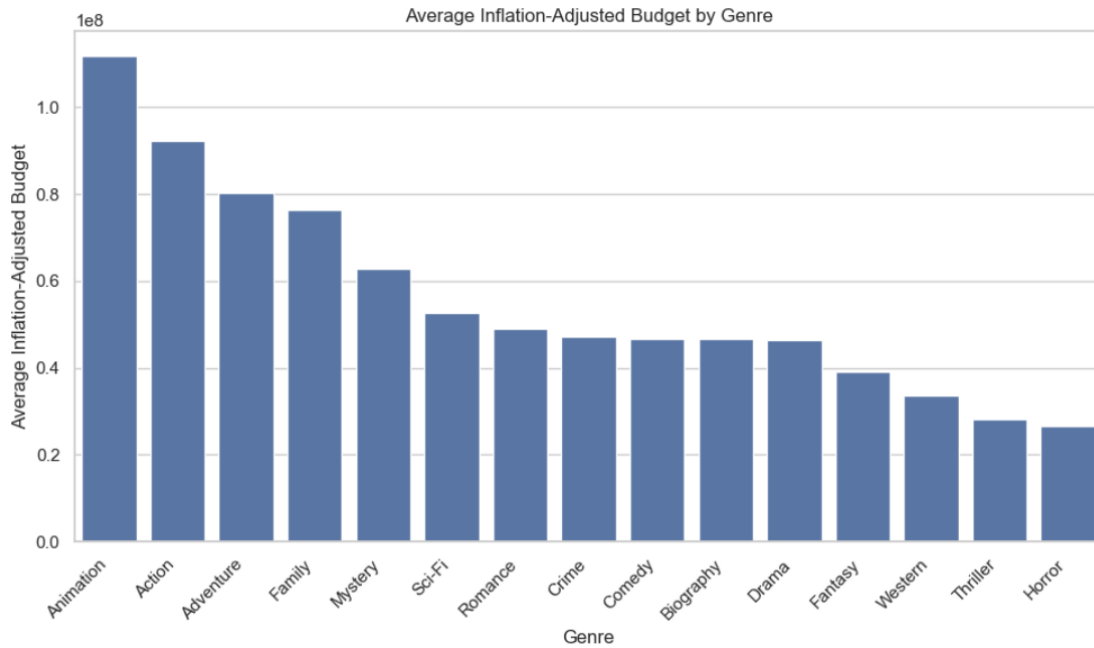
Average profits were calculated by grouping movies according to their country of production. Malta ranked as the most profitable film-producing country, followed by Finland, South Africa, Taiwan, and Lebanon. Some of these countries ranked highly due to individual standout movies, while others benefit from strong international distribution networks.

Table 2. Top 10 Countries by Average Profit

Rank	Country	Average Profit (USD)
1	Malta	139,781,558.52
2	Finland	104,776,205.57
3	South Africa	67,088,799.20
4	Taiwan	61,739,242.93
5	Lebanon	59,467,338.43
6	China	56,909,366.68
7	New Zealand	45,643,607.32
8	United Arab Emirates	31,169,300.96
9	Japan	29,800,587.34
10	Spain	27,702,871.92

Q3. Which genres are more expensive to make?

Figure 1. Average Inflation-Adjusted Budget by Genre

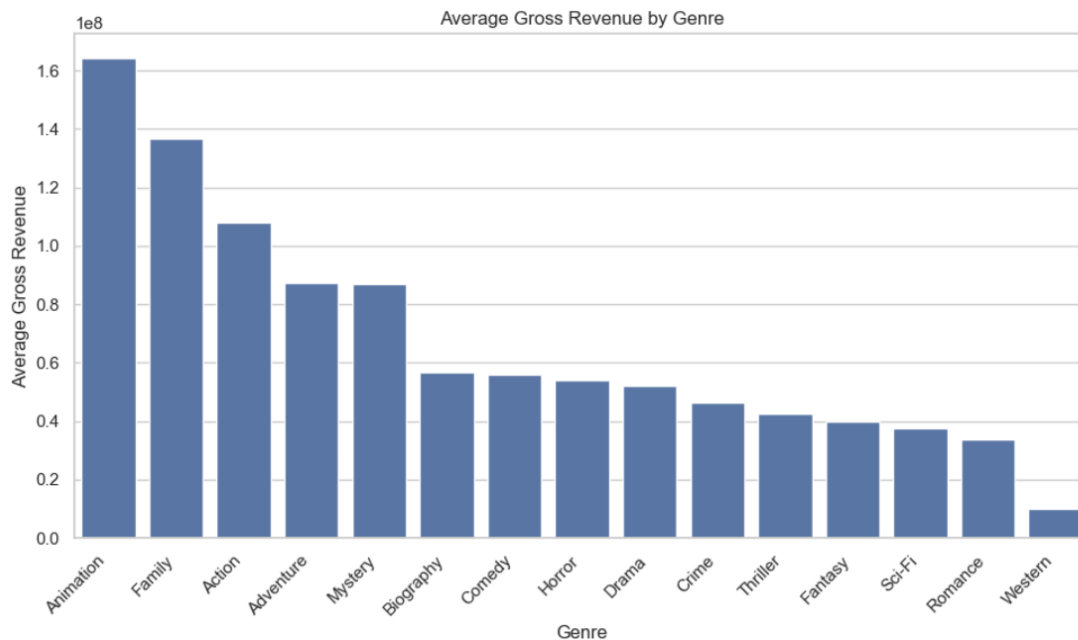


Interpretation:

Sci-Fi, Adventure, and Action films have the highest inflation-adjusted budgets. Genres like Comedy and Horror require significantly smaller budgets.

Q4. Which genres produce higher gross revenues?

Figure 2. Average Gross Revenue by Genre

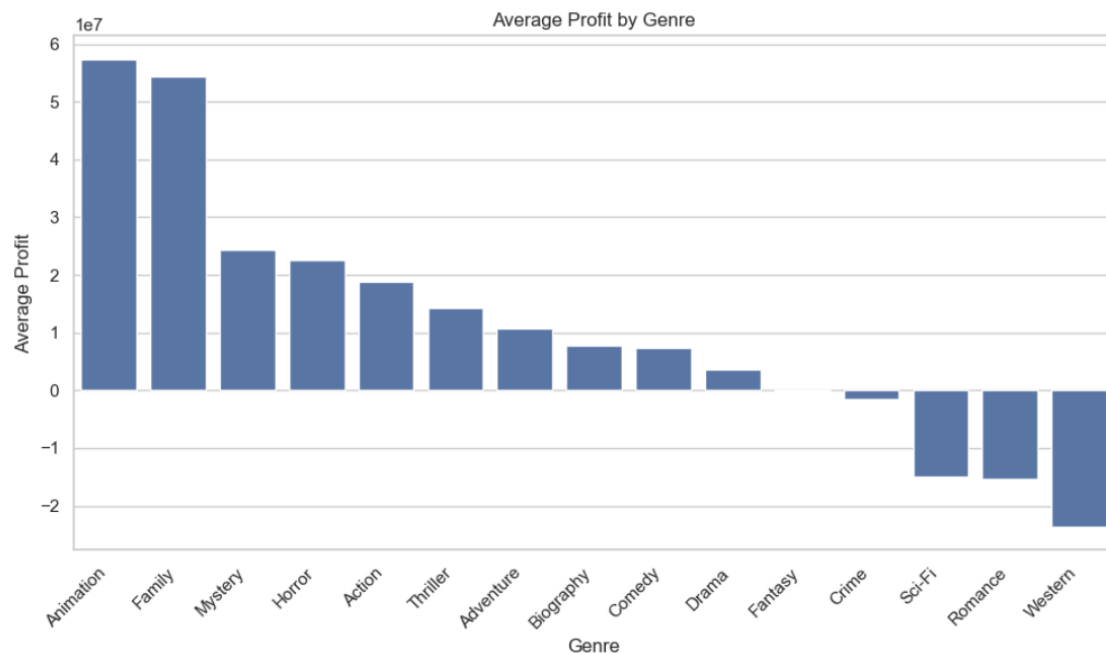


Interpretation:

Action and Adventure movies earn the highest gross revenue due to their broad audience appeal and strong international markets.

Q5. Which genres generate the most profit?

Figure 3. Average Profit by Genre

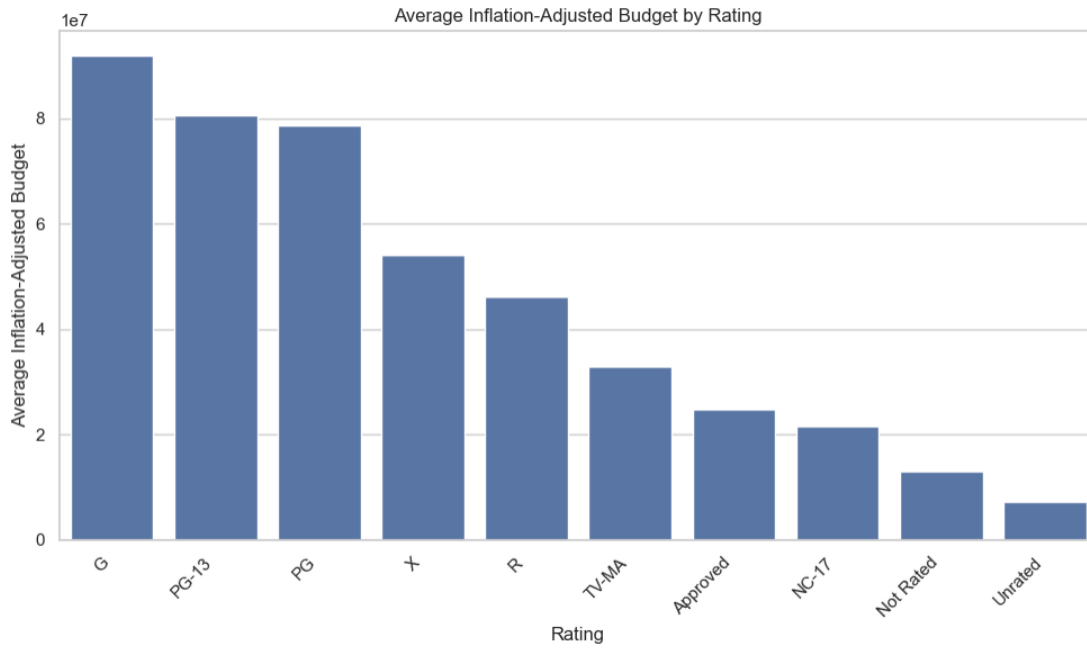


Interpretation:

Animation and Family movies generate notably high profits, while Western and Romance genres perform the weakest.

Q6. Which ratings are more expensive to make?

Figure 4. Average Inflation-Adjusted Budget by Rating

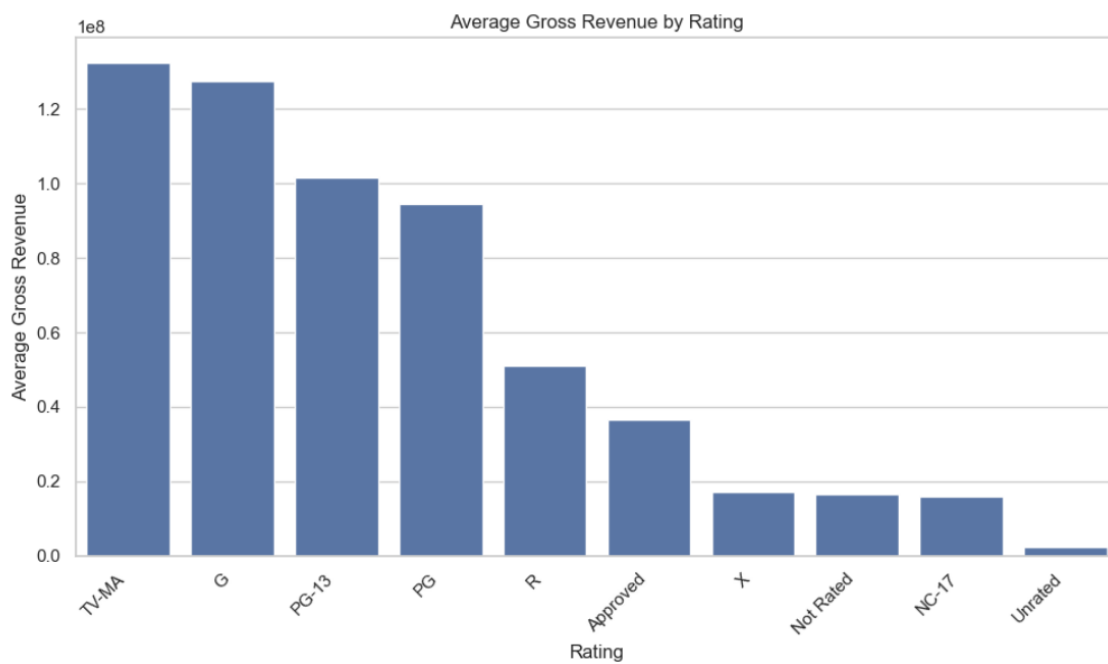


Interpretation:

PG-13 and R-rated movies carry the largest budgets, while G-rated films and TV-MA content require far less investment.

Q7. Which ratings produce higher gross revenues?

Figure 5. Average Gross Revenue by Rating

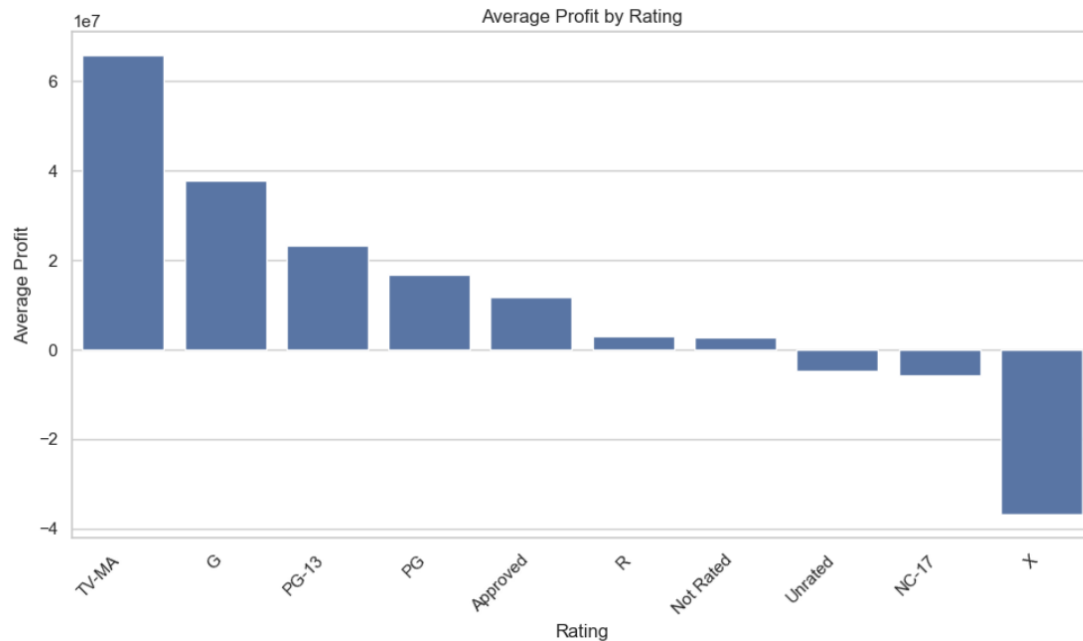


Interpretation:

PG-13 movies earn the highest gross revenue, followed by R-rated films.

Q8. Which ratings generate the most profit?

Figure 6. Average Profit by Rating

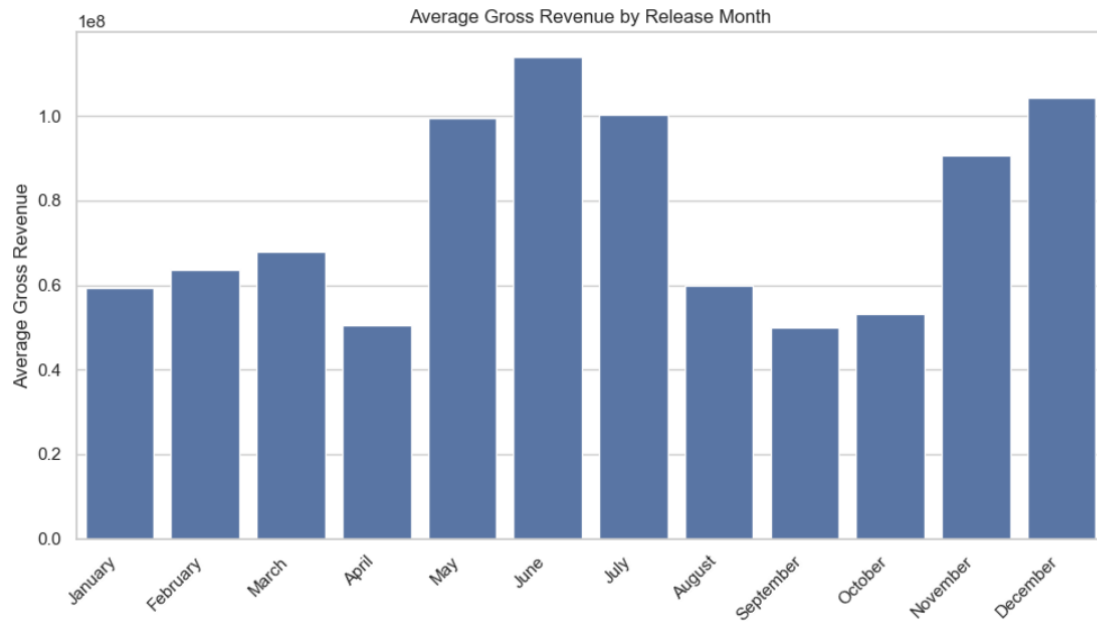


Interpretation:

G-rated and TV-MA films produce the highest profits. NC-17 and X-rated films perform poorly.

Q9. Which months lead to higher gross revenues?

Figure 7. Average Gross Revenue by Release Month

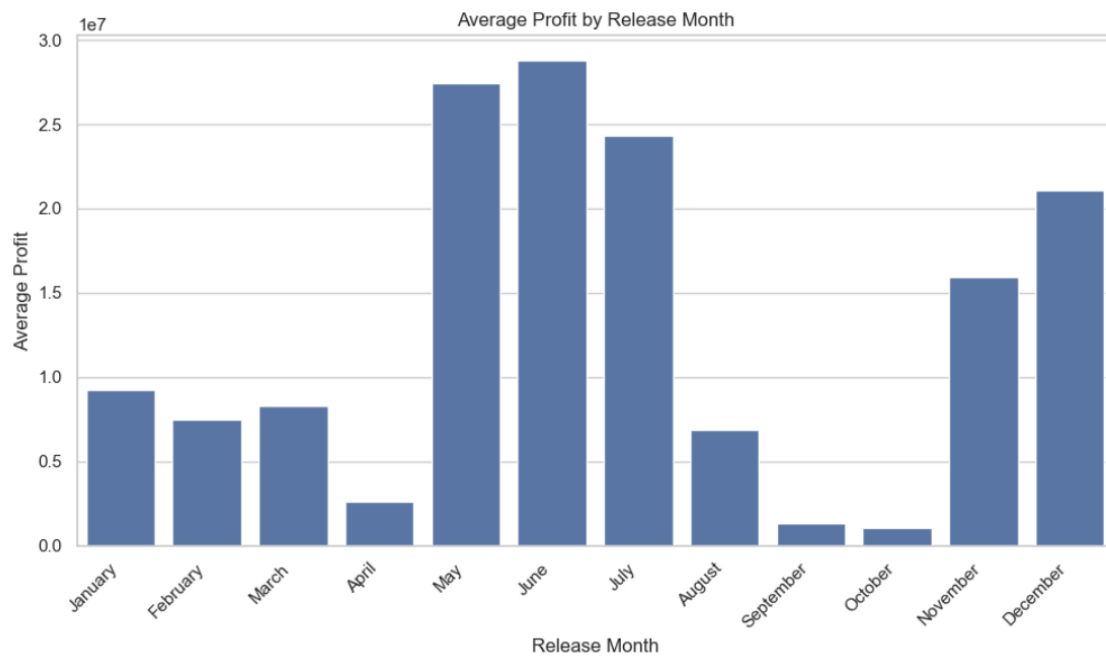


Interpretation:

June, July, and December perform the strongest. September and October are the lowest-revenue months.

Q10. Which months lead to higher profits?

Figure 8. Average Profit by Release Month

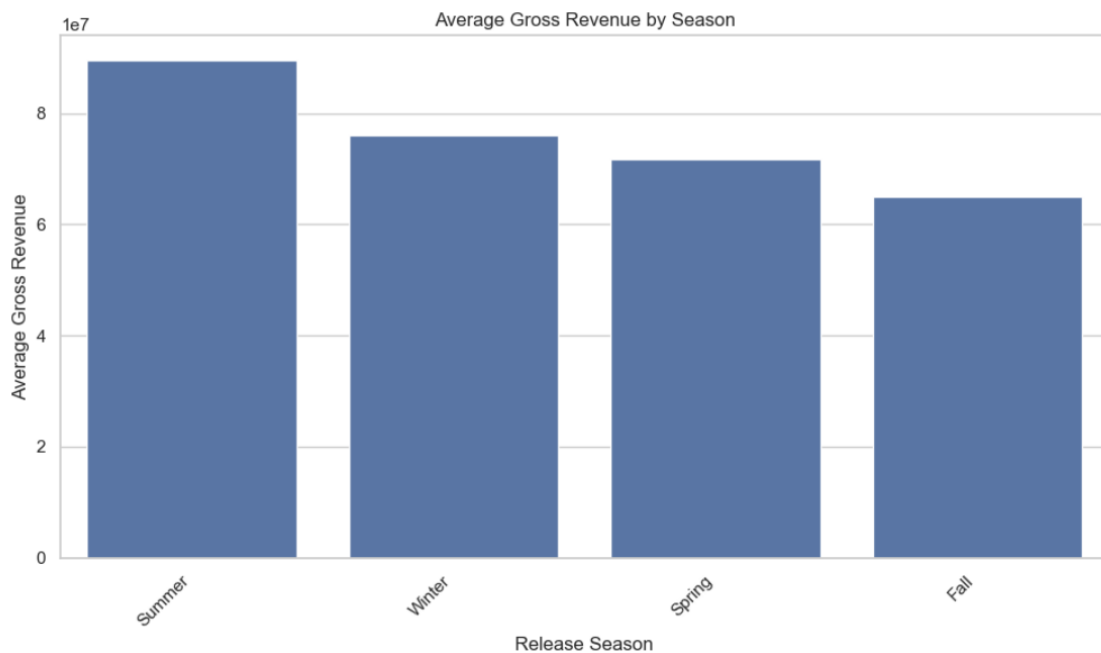


Interpretation:

May–July and December produce the highest profits. Early fall (September–October) shows the lowest profitability.

Q11. Which seasons lead to higher gross revenue?

Figure 9. Average Gross Revenue by Season

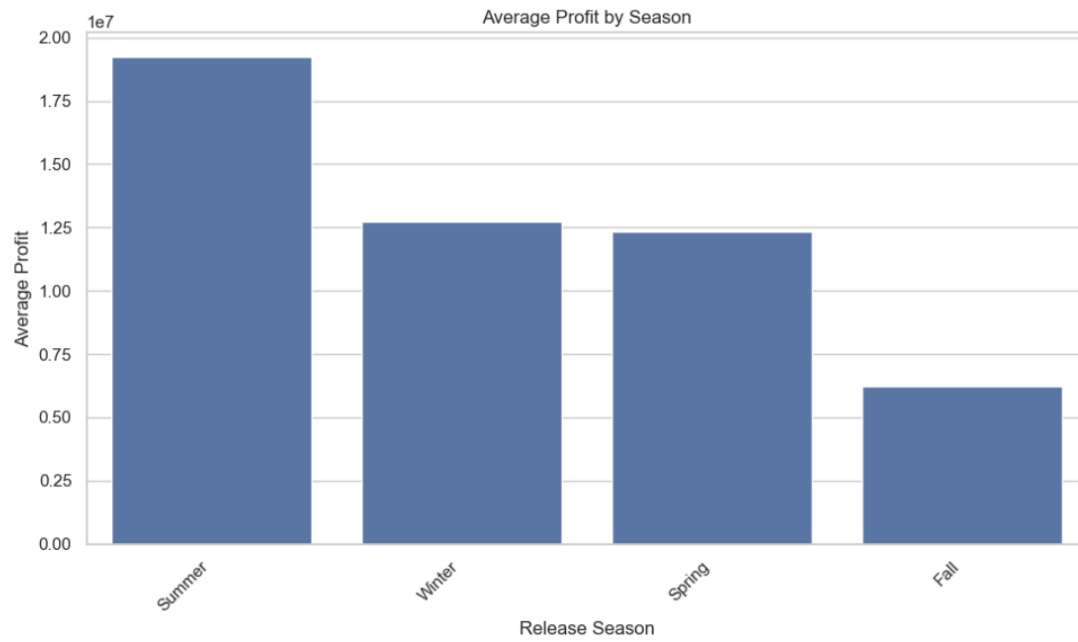


Interpretation:

Summer is the strongest season for gross revenue, followed by Winter.

Q12. Which seasons lead to higher profits?

Figure 10. Average Profit by Season

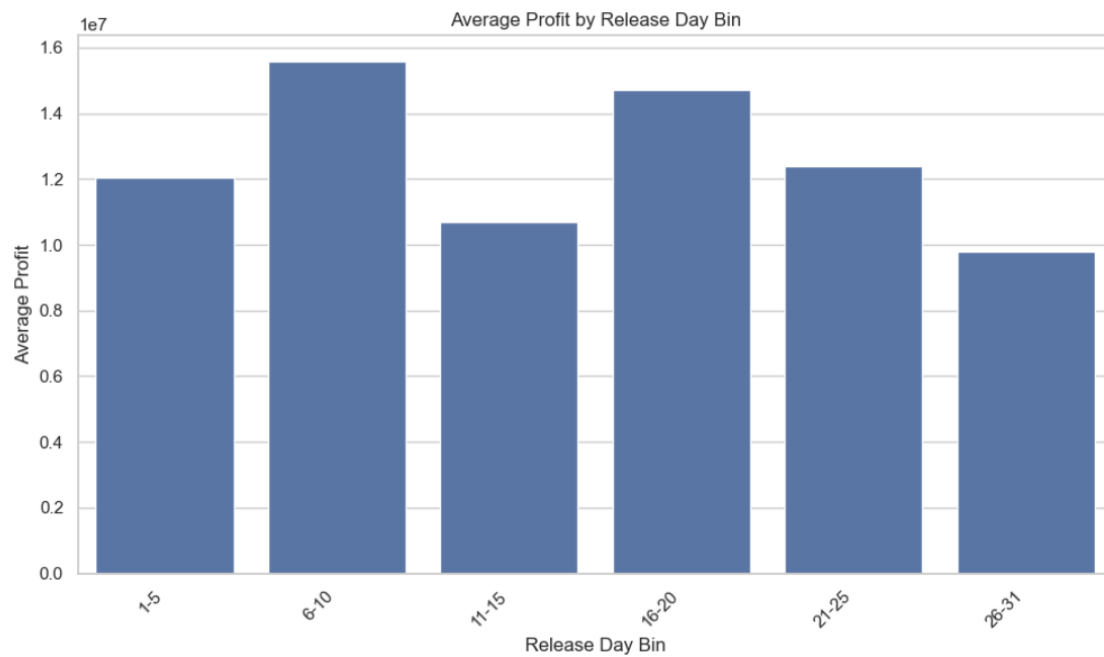


Interpretation:

Summer again leads in profitability; Fall is consistently the weakest season.

Q13. Which release-day bins lead to the highest profit?

Figure 11. Average Profit by Release Day Bin



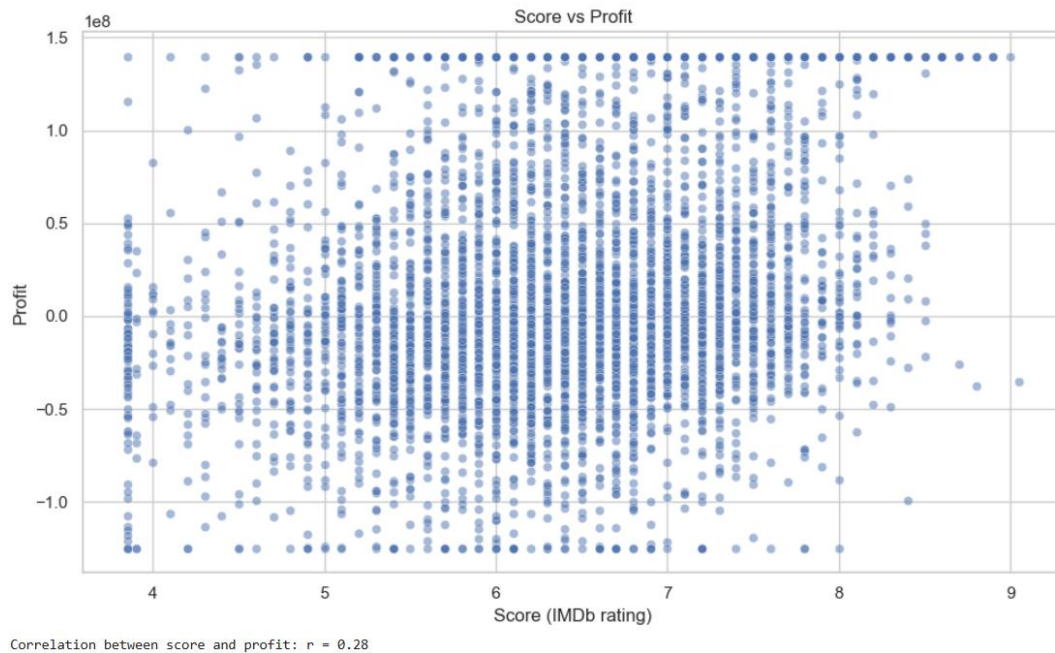
Interpretation:

Movies released between 6–10 generate the highest profits.

Releases in 26–31 perform the worst.

Q14. What is the relationship between score and profit?

Figure 12. Score vs. Profit (Scatterplot)



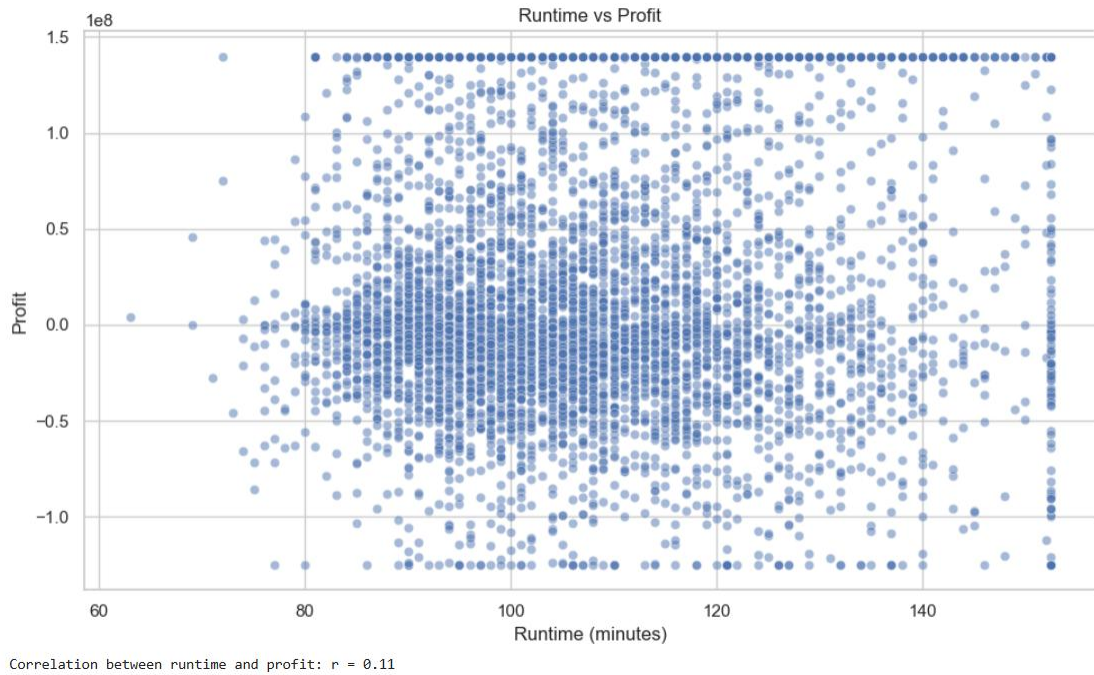
Correlation: $r = 0.28$ (Low)

Interpretation:

Higher IMDb scores correlate weakly with higher profits.

Q15. What is the relationship between runtime and profit?

Figure 13. Runtime vs. Profit (Scatterplot)



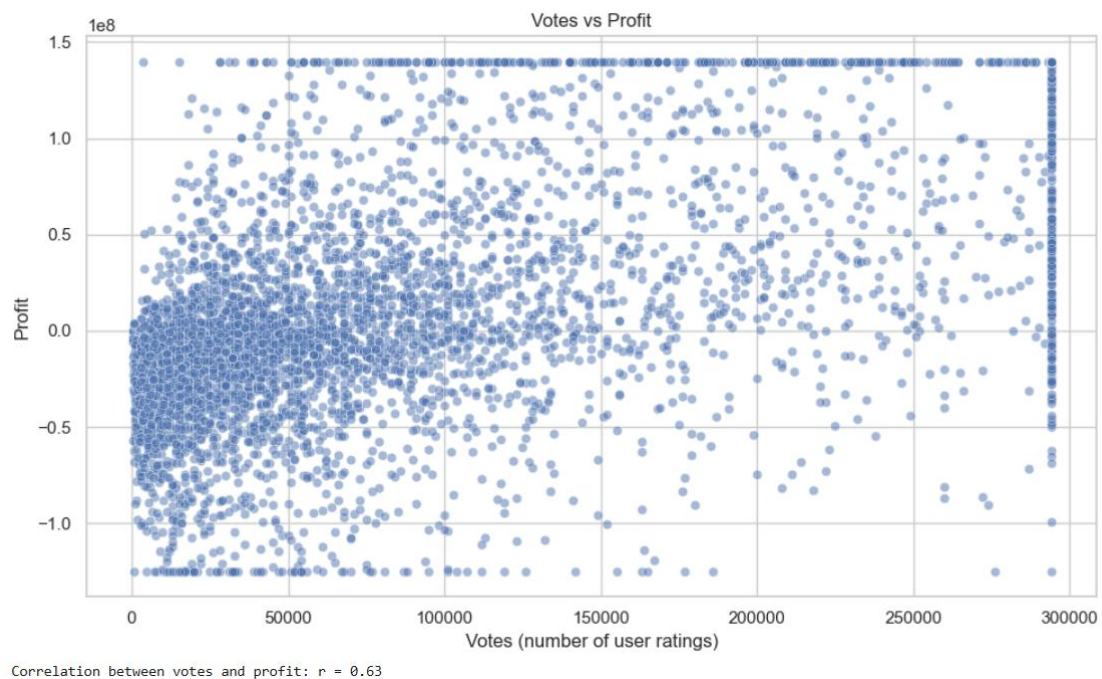
Correlation: $r = 0.11$ (Low)

Interpretation:

Runtime does not meaningfully influence profitability.

Q16. What is the relationship between votes and profit?

Figure 14. Votes vs. Profit (Scatterplot)



Correlation: $r = 0.63$ (High)

Interpretation:

Votes are the strongest predictor of profit among descriptive variables.

Q17. What was the relationship between votes and profit?

There is a strong positive relationship between the number of votes a movie receives and its profit. The correlation coefficient is $r = 0.63$, indicating a high correlation. This means that movies with higher audience engagement—reflected by a larger number of user votes—tend to generate substantially higher profits. The scatterplot also supports this trend, showing that as votes increase, profit rises sharply. Therefore, audience engagement is one of the strongest descriptive indicators of financial success.

Q18. Based on your findings, what would you recommend to a movie producer who wants to maximize gross revenues?

To maximize gross revenue, a movie producer should strategically align several factors identified in the analysis:

- Choose high-revenue genres, such as *Action*, *Adventure*, and *Animation*, which consistently generate the largest gross earnings.
- Release the movie during peak months, especially *June*, *July*, and *December*, when audience turnout is highest.
- Use a PG-13 rating, as it appeals to the broadest audience and produces the highest gross revenues across rating categories.
- Focus on strong marketing strategies to increase audience engagement, since high vote counts correlate with greater box-office performance.
- Consider Summer or Winter releases, which historically deliver the strongest box-office results.

These strategies maximize visibility, audience size, and commercial performance.

Q19. Based on your findings, what would you recommend to a movie producer who wants to maximize profit?

To maximize profit rather than just revenue, a producer should optimize both earnings and cost efficiency:

- Select highly profitable genres, such as *Animation*, *Family*, and *Comedy*, which consistently produce strong profits.
- Avoid low-profit genres, such as *Romance*, *Western*, and *Certain Sci-Fi films*.

- Release movies during the most profitable periods, particularly *May–July* and *December*, which yield the strongest profits.
- Control production costs, especially when working in high-budget genres (Action, Sci-Fi), which may generate high revenue but lower profit margins.
- Maximize audience engagement, as profits rise dramatically when vote counts increase.
- Avoid restrictive ratings (e.g., NC-17 or X), which reduce audience size and profitability.

Overall, strategic genre selection, cost management, and timing play crucial roles in maximizing financial returns.

Part C – Hypothesis Testing (Multiple Linear Regression)

To test hypotheses H1–H9, we estimated a multiple linear regression model with **profit** as the dependent variable and the following predictors:

- **Continuous variables:**
 - Score (H₁)
 - Votes (H₂)
 - Years since production (H₃)
 - Movie name length (H₇)
 - Runtime (H₈)
- **Categorical variables (dummy coded):**
 - Rating (H₄)
 - Genre (H₅)
 - Month of release (H₆)
 - Release day bin (H₉)

For the categorical variables, we created dummy variables and dropped one reference category from each group as required: “**Unrated**” for rating, “**Biography**” for genre, and “**26–31**” for release_day_bin. The regression was run twice: once using the original (unstandardized) predictors and once using standardized predictors (after scaling all independent variables). The standardized model provides the standardized betas that are used to compare relative effect sizes.

Model Fit

The unstandardized regression model achieved an **R² of 0.464** and an **Adjusted R² of 0.460**. This means that approximately **46% of the variation in movie profit** is explained by the set of predictors in the model. Given the complexity and noise of the movie industry, this level of explanatory power is reasonably strong for a cross-sectional dataset. The overall F-statistic was significant ($F(44, 5378) = 106.0$, $p < .001$), indicating that the model as a whole provides a significantly better fit than a model with no predictors.

We also checked multicollinearity using VIF. Some release-month dummy variables showed very high VIF values, reflecting strong correlations among month dummies. However, because the goal of this assignment is hypothesis testing (rather than building a parsimonious forecasting model), we retained all required variables and interpreted the results with this limitation in mind.

Regression Results and Hypothesis Summary

The table below summarizes the key results at the **construct level** (one row per hypothesis). For categorical variables, we report the most relevant significant categories compared to their reference groups.

Independent Variable (Construct)	Example Coefficients (vs. baseline)	Unstandardized Beta (Direction)	Standardized Beta (Direction)	p-value	Hypothesis Supported ?
Score (H ₁)	Continuous (per 1-point increase)	$\beta = 2,064,619.28$	$\beta^* = 1,921,952.01$	0.026	Yes
Votes (H ₂)	Continuous (per 1 vote)	$\beta = 402.54$	$\beta^* = 37,304,302.08$	$< .001$	Yes
Years since production (H ₃)	Continuous (per additional year since release)	$\beta = -1,109,874.27$	$\beta^* = -11,714,838.39$	$< .001$	Yes
Movie name length (H ₇)	Continuous (per extra character in the title)	$\beta = 478,168.54$	$\beta^* = 3,821,891.00$	$< .001$	Yes
Runtime (H ₈)	Continuous (per additional minute)	$\beta = -327,573.42$	$\beta^* = -5,424,040.39$	$< .001$	Yes
Rating (H ₄)	e.g., rating_TV-MA vs. Unrated (baseline)	Most rating dummies not significant (e.g., TV-MA $\beta = 60,159,030.85$, $p = 0.086$)	Small, non-significant effects overall	Most $p > .05$	No (overall)

Genre (H ₅)	Animation, Comedy, Horror vs. Biography (baseline)	Animation β = 16,574,673.03, Comedy β = 8,501,722.28, Horror β = 22,217,xxx (all positive)	Animation β^* $\approx$ 3.65M, Comedy β^* $\approx$ 3.80M, Horror β^* $\approx$ 4.67M	$p < .01$ for Animation, Comedy, Horror	Yes (partially)
Month of release (H ₆)	May, July, October vs. baseline month (one month omitted as reference)	May β = 9,173,805.57 ($p < .001$), July β = 5,299,111.77 ($p = 0.027$), October β = -4,846,953.65 ($p = 0.031$)	Positive for May & July, negative for October	Several $p < .05$	Yes
Release day bin (H ₉)	1–5, 6–10, 11–15, 16–20, 21–25 vs. 26–31 (baseline)	All release_day_bin coefficients small and non-significant	Small, non-significant	All $p > .05$	No

Note: Exact horror coefficient from the full output (genre_Horror) was $\beta \approx 22.22\text{M}$ with $p < .001$, making it one of the strongest positive genre effects.

Interpretation of Key Predictors

- H₁ – Score:**
 Score has a positive and statistically significant effect on profit ($\beta = 2.06\text{M}$, $p = 0.026$). Holding all other variables constant, a 1-point increase in IMDb score is associated with about \$2.06 million higher profit. This supports H₁ and suggests that better-rated movies tend to be more profitable, although the effect is modest compared to votes.
- H₂ – Votes (audience engagement):**
 Votes are a strong positive predictor of profit ($\beta = 402.54$, $p < .001$). For every additional 10,000 votes, profit increases by roughly \$4.0 million ($10,000 \times 402.54$). The standardized beta for votes is the largest in the model, indicating that audience engagement is the single strongest predictor of profitability.
- H₃ – Years since production:**
 Years since release has a negative and significant relationship with profit ($\beta = -1.11\text{M}$, $p < .001$). Each additional year since release is associated with about \$1.11 million lower profit, controlling for other variables. This supports H₃ and suggests that more recent films tend to generate higher profits in this dataset.
- H₇ – Movie name length:**
 Movie name length is positively and significantly related to profit ($\beta = 478,168.54$, $p < .001$). Each extra character in the title is associated with nearly \$0.48 million higher

profit on average. While the effect size per character is moderate, the standardized beta indicates that title length is a meaningful (though secondary) predictor compared to votes and recency.

- **H₈ – Runtime:**
Runtime shows a negative and significant effect ($\beta = -327,573.42$, $p < .001$). Longer movies tend to earn slightly lower profit, holding other factors constant. The magnitude is smaller than votes or years since release, but H₈ is supported: runtime does affect profitability, and in this sample, longer runtimes are associated with reduced profit.
- **H₄ – Rating category:**
After controlling for all other variables, most rating dummies (G, PG, PG-13, R, TV-MA, etc.) are not statistically significant (all $p > .05$). This means we do not find strong evidence that rating categories, by themselves, drive profit differences once genre, score, votes, and timing are included. Thus, H₄ is not supported overall.
- **H₅ – Genre:**
Genre has a meaningful and partially significant effect on profit. In particular, Animation, Comedy, and Horror films show significantly higher profits compared to the Biography baseline (Animation $\beta \approx 16.57\text{M}$, Comedy $\beta \approx 8.50\text{M}$, Horror $\beta \approx 22.22\text{M}$, all $p < .01$). Other genres (Action, Drama, Romance, etc.) are not significant. Overall, this supports H₅ in the sense that some genres clearly outperform others financially.
- **H₆ – Month of release:**
Several months show significant coefficients: May ($\beta \approx +9.17\text{M}$, $p < .001$) and July ($\beta \approx +5.30\text{M}$, $p = 0.027$) are associated with higher profits, while October shows a negative effect ($\beta \approx -4.85\text{M}$, $p = 0.031$) relative to the reference month. This pattern supports H₆ and suggests that summer and early-summer releases (especially May and July) are financially advantageous, whereas October releases underperform.
- **H₉ – Release day of month:**
The release_day_bin variables (1–5, 6–10, 11–15, 16–20, 21–25 vs. 26–31) are all non-significant (all $p > .05$). This indicates that, after controlling for genre, month, score, votes, and other factors, the day group within the month does not meaningfully affect profit. Therefore, H₉ is not supported.

Relative Importance (Standardized Betas)

Using the standardized regression model, the two strongest predictors in absolute terms are:

1. **Votes** – the largest positive standardized beta, indicating that audience engagement is the most influential driver of profit.
2. **Years since production** – the largest negative standardized beta, showing that recency (being a newer film) is the second most powerful predictor, with older films earning substantially less profit on average.

Genres such as Horror, Animation, and Comedy also have sizeable positive standardized betas, indicating that genre choice meaningfully shifts profitability once other factors are controlled.

Part D — Predictive Modeling

A movie director is preparing three new films and wants to estimate their potential profitability. To support this decision, we built two machine learning models—a decision tree and a neural network—to predict profit using a 70–30 train-test split. Only appropriate input variables were included in the models, and **budget and gross were excluded**, as they are directly tied to profit and would violate predictive validity. Missing values for the new movies' *score* and *votes* were replaced with the dataset's mean values, as instructed.

Movie	Rating	Genre	Country	Budget (USD)	Runtime (min)
Gorga	R	Drama	United States	130,000,000	250
King Wolf	R	Drama	United States	250,000,000	135
Hellish Gray	PG	Comedy	United States	310,000,000	108

Model Evaluation

1. Decision Tree Model

After tuning hyperparameters (`max_depth`, `min_samples_split`, `min_samples_leaf`), the best model used:

- `max_depth = 7`
- `min_samples_leaf = 10`
- `min_samples_split = 50`

Test-Set Performance:

- $R^2 = 0.522$
- $RMSE = \$43,401,156.83$

Interpretation:

The decision tree explains 52.2% of the variation in profits and has an RMSE of \$43 million, indicating moderate predictive strength. The model captures non-linear relationships effectively and performs well given the natural volatility of movie profits.

2. Neural Network (MLP) Model

Hyperparameter tuning yielded the following best configuration:

- Hidden layers = (32, 16)
- Activation = ReLU

- $\alpha = 0.01$

Test-Set Performance:

- $R^2 = 0.463$
- $RMSE = \$45,980,049.09$

Interpretation:

The neural network explains 46.3% of the profit variance, slightly lower than the decision tree. The higher RMSE also indicates reduced accuracy compared to the tree. Although neural networks can model more complex interactions, the available feature set favors simpler models.

3. Comparison of Models

Model	Test R^2	Test RMSE
Decision Tree	0.522	\$43.40M
Neural Network	0.463	\$45.98M

Conclusion:

The Decision Tree Regressor is the best-performing model and is therefore used for final predictions.

4. Predicted Profit for Upcoming Movies

Using the decision tree model and imputing missing *score* and *votes* with dataset means:

Movie	Predicted Profit (USD)
Gorga	\$69,065,897.70
King Wolf	\$69,065,897.70
Hellish Gray	\$69,065,897.70

References:

[Kaggle: Your Machine Learning and Data Science Community](#)